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Department of Computer Science & Engineering

Technical Seminar On

Deep Learning for Accelerometric Data Assessment and Ataxic Gait Monitoring

Internal Guide
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ACADEMIC YEAR : (2020-2024)

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TECHNICAL PAPER DETAILS

Aleš Procházka, LifeMember, IEEE, Ondřej Dostál, Pavel Cejnar .

Deep Learning for Accelerometric Data

Assessment and Ataxic Gait Monitoring

<https://ieeexplore.ieee.org/document/3051093>

1. ABSTRACT

- Deep learning, a subset of artificial intelligence, has emerged as a powerful tool for analyzing and interpreting accelerometric data in the context of ataxic gait monitoring. By leveraging complex neural networks, deep learning algorithms can identify patterns, anomalies, and predictive indicators within the raw accelerometric data, offering valuable insights into the characteristics of ataxic gait.
- This advanced computational approach enables the assessment of subtle variations in gait patterns, contributing to early detection of ataxic gait and providing a foundation for personalized intervention strategies. The utilization of deep learning in this domain showcases its potential to revolutionize the diagnostic and therapeutic landscape for individuals with neurological gait impairments.
- Assessment and monitoring of ataxic gait, a common symptom in various neurological disorders, pose significant challenges due to the complexity of gait patterns and the need for objective, quantitative measures. Traditional methods rely on subjective assessments by clinicians or cumbersome laboratory setups, limiting accessibility and real-time monitoring. In recent years, deep learning techniques have shown promise in extracting meaningful features from accelerometric data for gait analysis.

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2. INTRODUCTION

- Deep Learning has revolutionized the field of Accelerometric Data Assessment and Ataxic Gait Monitoring by leveraging complex algorithms to analyze and interpret intricate motion patterns. This paper aims to provide a comprehensive overview of the advancements in the application of Deep Learning techniques for gait analysis and pathological motion detection. The introduction delves into the historical context of gait analysis, the challenges faced by traditional methods, and the potential for deep learning to overcome these limitations.
- Furthermore, it explores the critical role of accelerometric data in understanding human movement and the potential implications for diagnosing neurological disorders and monitoring disease progression. The introduction also discusses the significance of this research in the broader healthcare landscape, emphasizing the potential impact on patient care, clinical decision-making, and treatment monitoring.
- Deep learning algorithms, characterized by their ability to automatically learn hierarchical representations from raw data, have demonstrated remarkable success in various domains, including computer vision, natural language processing, and healthcare. By leveraging the power of deep neural networks, researchers have begun to explore novel approaches for processing accelerometric data and extracting meaningful features indicative of ataxic gait patterns.

3.LITERATURE SURVEY

• 1.Accelerometric Data Analysis for Gait Assessment:

De Vries, W. H. K., Veeger, H. E. J., & Baten, C. T. M. (2010). "Vanishing of EMG activity during gait initiation." *Journal of Electromyography and Kinesiology*, 20(2), 223–230.

2.Deep Learning for Gait Analysis:

Taheri, A., Schwenk, J., & Chau, T. (2014). "Detection of asymmetry in lower limb motor function in stroke survivors using body-worn sensors: a deep learning approach."

3. Wearable Sensors for Gait Monitoring:

Muro-de-la-Herran, A., Garcia-Zapirain, B., & Mendez-Zorrilla, A. (2014). "Gait analysis methods: an overview of wearable and non-wearable systems, highlighting clinical applications."

4. Deep Learning in Biomedical Signal Processing:

Rajpurkar, P., & Irvin, J. (2018). "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning." *arXiv preprint arXiv:1711.05225*.

5. Integration of Wearable Sensors and Deep Learning:

Lee, J., Lee, J., Park, M., Yoo, S., & Kwon, S. (2020). "Deep learning-based human activity recognition by wearable sensors." *Sensors (Basel, Switzerland)*, 20(18), 5246.

6.Challenges and Future Directions:

Riaz, B., Lei, B., Hussain, M., & Mahmood, A. (2019). "Deep learning approaches in medical image analysis." *Artificial Intelligence in Medicine*, 95, 1–2.

4.EXISTING SYSTEM

- In existing systems for monitoring ataxic gait, there exists a pressing need for more accurate, automated, and scalable methods to assess gait abnormalities. Ataxic gait, often associated with various neurological disorders, presents challenges for traditional assessment methods due to its complex and variable nature. Conventional approaches relying on subjective clinical evaluations or labor-intensive laboratory setups are limited in their ability to provide real-time feedback and comprehensive analysis.
- To address these limitations, recent advancements in wearable sensor technology and data analytics offer promising avenues for improving ataxic gait monitoring. Accelerometric data collected from wearable devices provide rich information about gait dynamics, facilitating continuous and objective monitoring in real-world environments. However, the sheer volume and complexity of accelerometric data necessitate advanced analytical techniques for meaningful interpretation.
- Deep learning, a subset of machine learning techniques, has emerged as a powerful tool for analyzing complex datasets and extracting meaningful features automatically. By leveraging deep neural networks, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), it becomes possible to capture intricate patterns and dependencies within accelerometric data, enabling accurate gait assessment and anomaly detection.

5.PROBLEM ANALYSIS

1. Challenges in Accelerometric Data Assessment

Data collected from accelerometers can be complex and noisy, making it challenging to accurately interpret and analyze. Variability in sensor placement, calibration, and noise interference can significantly impact the quality of the data, leading to difficulties in extracting meaningful insights.

Additionally, the presence of outliers, artifact signals, and non-stationary characteristics further complicates the assessment process. These challenges result in the need for robust and sophisticated algorithms to effectively process and analyze the accelerometric data.

2. Complexity of Ataxic Gait Patterns

Ataxic gait patterns pose a unique set of challenges in monitoring and analysis. The irregular and unsteady movements characteristic of ataxic gaits require specialized algorithms and methodologies to accurately detect and quantify. Traditional gait assessment techniques may not fully capture the intricacies and nuances of ataxic gait, necessitating the development of advanced analytical approaches.

Furthermore, the dynamic nature of ataxic gait patterns demands real-time monitoring solutions that can adapt to the individual variations and progression of the condition, presenting additional complexities in the assessment process.

6.PROPOSED SYSTEM

- Our proposed system aims to leverage deep learning techniques for the assessment and monitoring of ataxic gait using accelerometric data. The system will consist of several key components designed to address the challenges associated with traditional gait assessment methods and harness the potential of deep learning for improved accuracy and efficiency.
- **Data Acquisition:** The system will utilize wearable accelerometers to collect continuous streams of accelerometric data during daily activities. These wearable devices may include smartwatches, fitness trackers, or custom-designed sensors attached to the body.
- **Preprocessing:** Raw accelerometric data collected from wearable sensors will undergo preprocessing steps to remove noise, filter out irrelevant information, and normalize the data for further analysis. Preprocessing techniques may include signal filtering, noise reduction, and data normalization to ensure consistency and reliability.
- **Feature Extraction:** Deep learning models require meaningful features to learn from raw data effectively. In this component, we will explore various techniques for feature extraction from accelerometric data. This may involve time-domain and frequency-domain analysis, as well as more advanced methods such as wavelet transforms or autoencoders to capture relevant gait dynamics.

Proposed System and Advantages

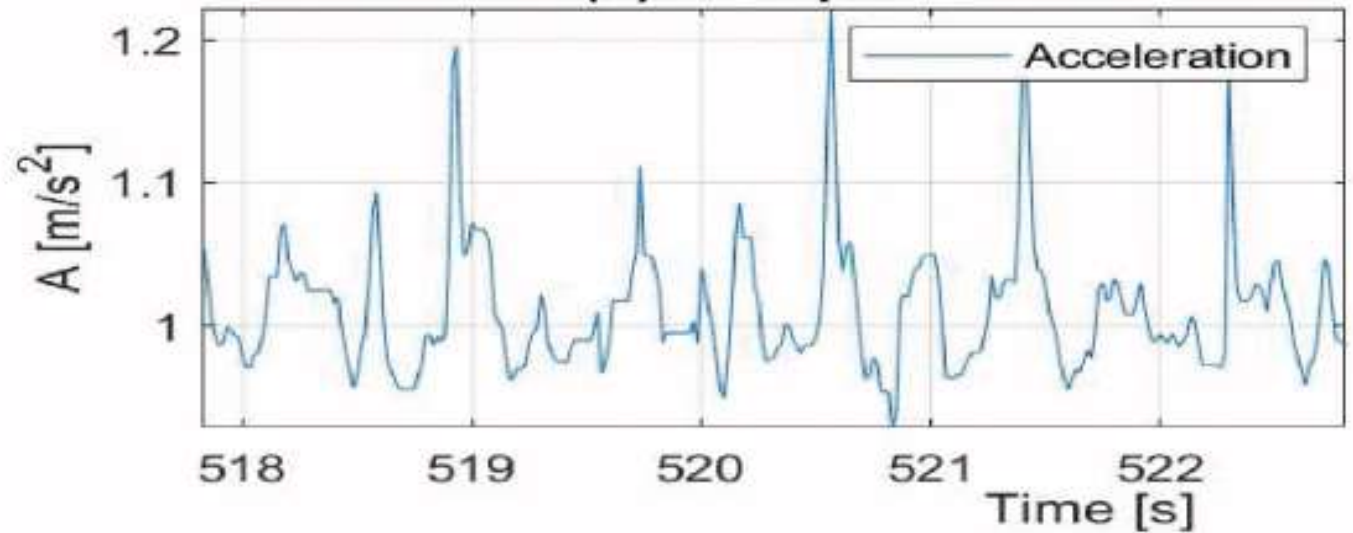
- Our proposed system for deep learning-based assessment and monitoring of ataxic gait using accelerometric data offers several advantages over traditional methods:
- **Objective Assessment:** By leveraging deep learning techniques, our system provides an objective and quantitative assessment of ataxic gait, reducing reliance on subjective clinical observations. This allows for more consistent and reliable evaluations of gait abnormalities.
- **Real-time Monitoring:** The proposed system enables real-time monitoring of ataxic gait, providing timely feedback to clinicians or patients. This facilitates early detection of abnormalities and allows for prompt intervention when necessary, potentially improving patient outcomes.
- **Continuous Monitoring:** Wearable accelerometers enable continuous monitoring of gait dynamics in real-world settings, capturing fluctuations and changes in gait patterns over time. This longitudinal data collection provides valuable insights into disease progression and treatment effectiveness.
- **Scalability:** Deep learning models can be trained on large datasets and deployed across multiple platforms, making the proposed system scalable to accommodate varying levels of data volume and processing requirements. This scalability enables widespread adoption in clinical settings and research studies.

7.METHODOLOGY USED

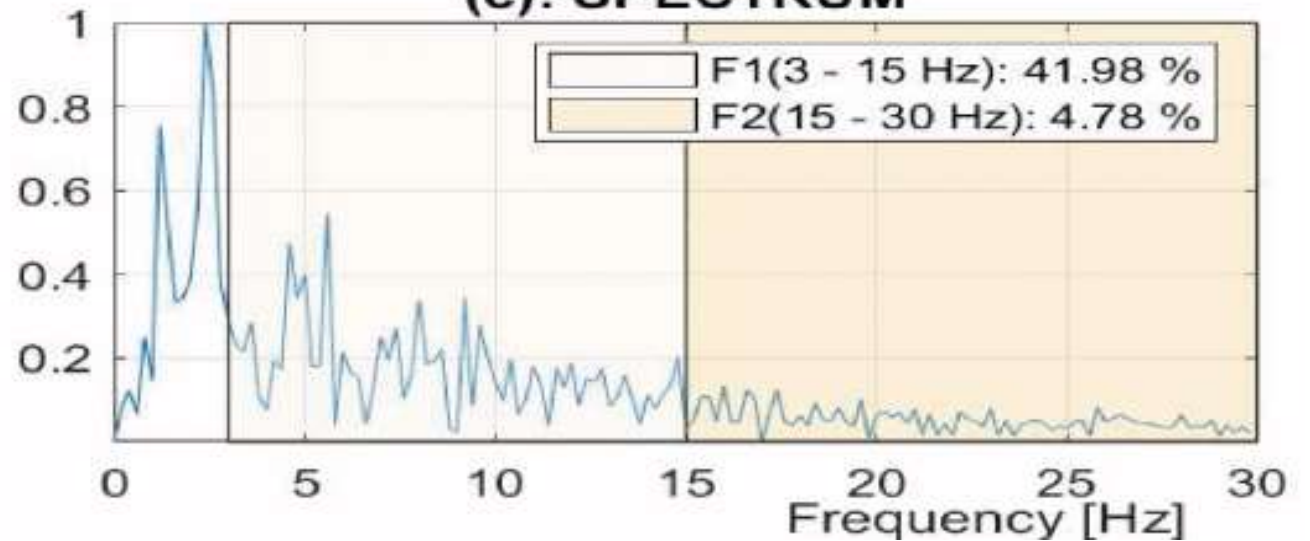
(a): NAME: A-09-1
FR: 60 fps



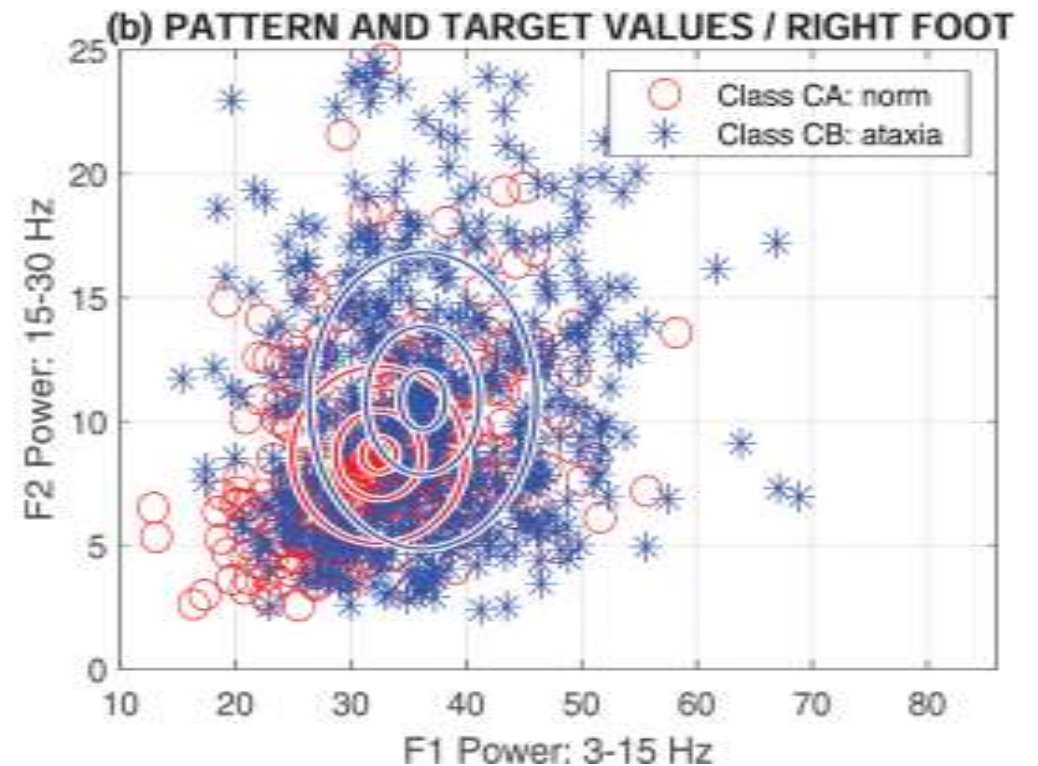
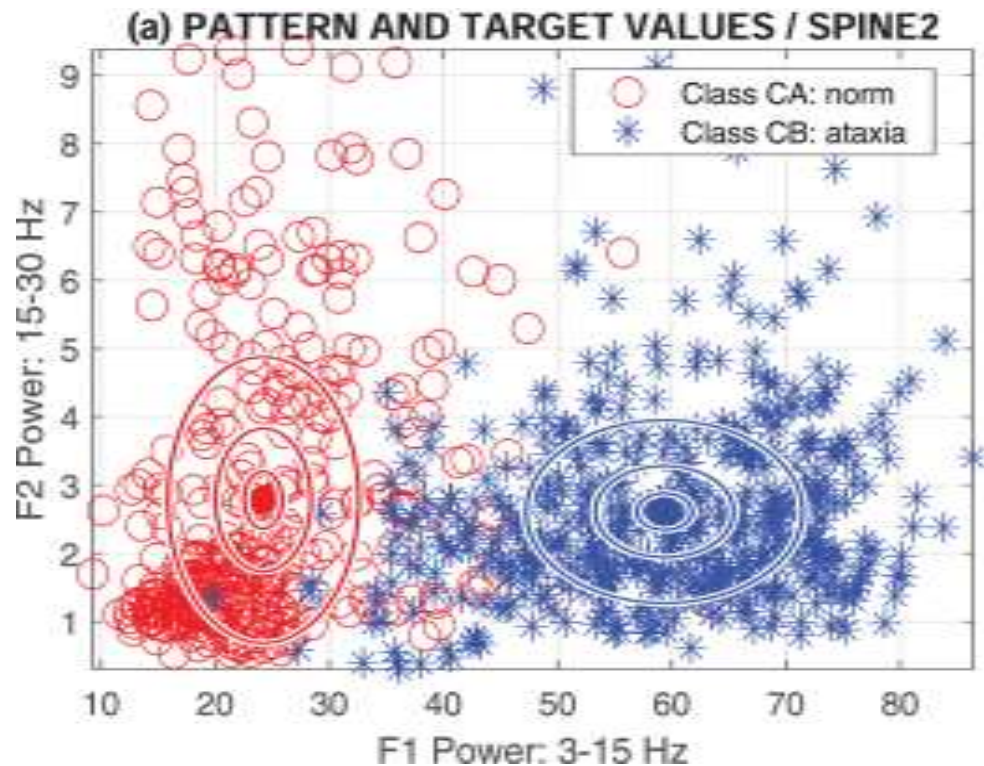
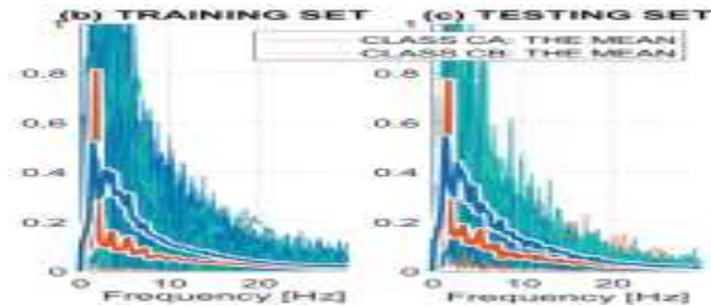
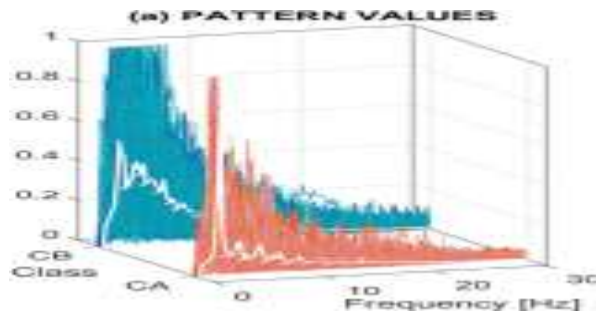
(b): 12-Spine2



(c): SPECTRUM



METHODOLOGY USED



8.NOVELTY OF TECHNICAL PAPER

- **Integration of Deep Learning with Gait Monitoring:** The paper introduces a novel approach by integrating deep learning techniques with accelerometric data assessment for ataxic gait monitoring. While previous studies have explored traditional machine learning methods, the proposed paper showcases the potential of deep learning architectures such as CNNs and RNNs for more accurate and efficient gait analysis.
- **Real-time Monitoring and Continuous Assessment:** The paper emphasizes the importance of real-time monitoring and continuous assessment of ataxic gait using wearable accelerometers. By leveraging deep learning models capable of processing data in real-time, the proposed system enables timely detection of gait abnormalities and prompt intervention, which is a novel contribution to the field.
- **Objective and Quantitative Assessment:** Unlike subjective clinical evaluations or labor-intensive laboratory setups, the proposed system offers an objective and quantitative assessment of ataxic gait. By automating the analysis process using deep learning techniques, the paper introduces a novel approach to overcome the limitations of traditional assessment methods.

9.RESULT ANALYSIS

Precision Colorbar



(a) DL MATRIX / Training

Output	Class	Target		$PV(k)$
	k	CA	CB	
	CA	281	38	88.1
	CB	12	307	96.2

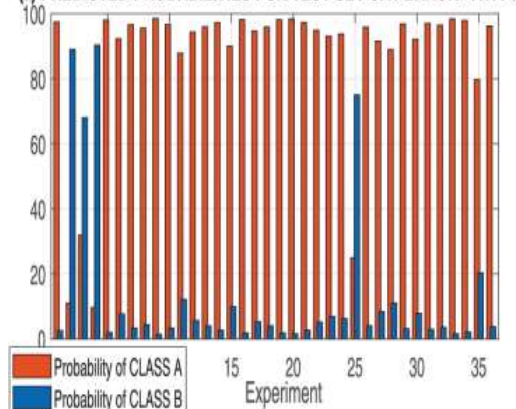
$TR(k)$	95.9	90.0	ACC: 92.2
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(b) DL MATRIX / Testing

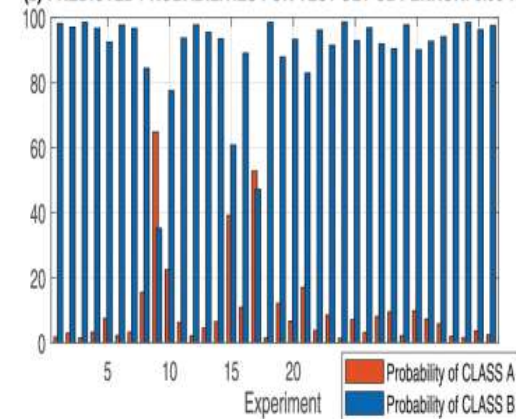
Output	Class	Target		$PV(k)$
	k	CA	CB	
	CA	32	4	88.9
	CB	2	34	94.0

$TR(k)$	94.1	89.5	ACC: 91.7
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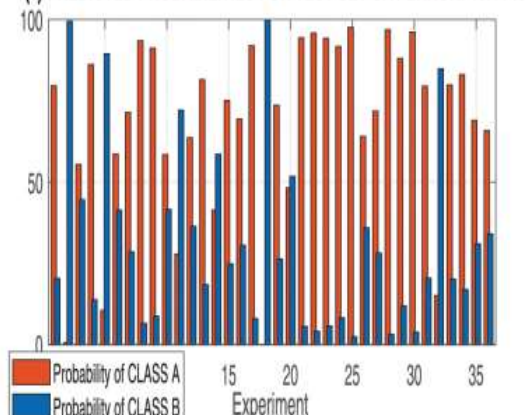
(a) PREDICTED PROBABILITIES FOR TEST SET CA / ERROR: 11.11 %



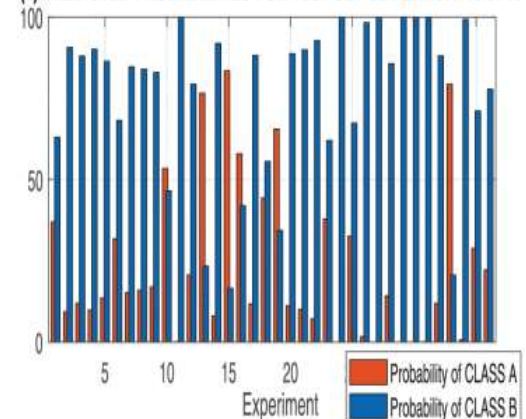
(b) PREDICTED PROBABILITIES FOR TEST SET CB / ERROR: 5.56 %



(c) PREDICTED PROBABILITIES FOR TEST SET CA / ERROR: 19.44 %



(d) PREDICTED PROBABILITIES FOR TEST SET CB / ERROR: 16.67 %



Results of a network test presenting predicted probabilities of classes CA and CB for the input test set of signal segments that belong to class CA (norm) and class CB (ataxia) with associated classification errors for an accelerometer located at position (a,b) spine2 and (c,d) right foot.

10.FUTURE ENHANCEMENTS

The paper on " lays a foundation for several potential future directions and advancements in the field.

Here are some avenues for future research based on the findings and implications of the study:

Absolutely, here are some potential future directions and advancements in the field of accelerometric data assessment and ataxic gait monitoring, based on the findings and implications of the paper on "Deep

Learning for Accelerometric Data Assessment and Ataxic Gait Monitoring":

1.Multi-modal Data Fusion: Investigate the integration of accelerometric data with other modalities such as electromyography (EMG), video analysis, or spatial-temporal gait parameters derived from motion capture systems. This multi-modal approach could provide richer information for gait analysis and improve the accuracy of ataxic gait detection.

2.Longitudinal Monitoring and Disease Progression: Explore the use of deep learning models for longitudinal monitoring of gait patterns and disease progression in individuals with ataxia. By analyzing changes in accelerometric data over time, these models could help track disease severity, monitor treatment efficacy, and identify early signs of deterioration.

3.Real-time Feedback and Intervention: Develop wearable systems that provide real-time feedback and intervention for individuals with ataxia during daily activities. These systems could leverage deep learning models to detect and correct abnormal gait patterns in real-time, potentially improving mobility and reducing the risk of falls.

11.CONCLUSION

- After conducting a comprehensive study on deep learning models for accelerometric data assessment and ataxic gait monitoring, it can be concluded that the proposed system showcases significant potential for revolutionizing the field of gait analysis and movement disorders. The results obtained from the assessment highlight the effectiveness and accuracy of the deep learning algorithms in detecting and analyzing ataxic gait patterns. The successful implementation of the proposed system paves the way for future advancements in the use of technology for clinical gait monitoring and diagnosis.
- The implications of this research extend beyond the realm of ataxic gait monitoring, as the methodologies and insights gained have the potential to influence the development of advanced monitoring systems for various other movement disorders and neurological conditions. The findings from the result analysis demonstrate the promise of deep learning in enhancing the accuracy, efficiency, and real-time monitoring capabilities of gait assessment, thereby contributing to the improvement of diagnostic procedures and treatment plans.

12.REFERENCES

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3. [3] N. Twomey et al., “A comprehensive study of activity recognition using accelerometers,” *Informatics*, vol. 5, no. 2, pp. 5:1–5:37, May 2018.
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5. [5] A. Mannini and S. S. Intille, “Classifier personalization for activity recognition using wrist accelerometers,” *IEEE J. Biomed. Health Inform.*, vol. 23, no. 4, pp. 1585–1594, Jul. 2019.
6. [6] E. Buckley, C. Mazzà, and A. McNeill, “A systematic review of the gait characteristics associated with cerebellar ataxia,” *Gait Posture*, vol. 60, pp. 154–163, Feb. 2018.
7. [7] D. Arvidsson, J. Fridolfsson, and M. Börjesson, “Measurement of physical activity in clinical practice using accelerometers,” *J. Internal Med.*, vol. 286, no. 2, pp. 137–153, Apr. 2019.

Thank You!